2019

Avocado Seedling Multiplication

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AVOCADO SEEDLINGS MOBILE APP

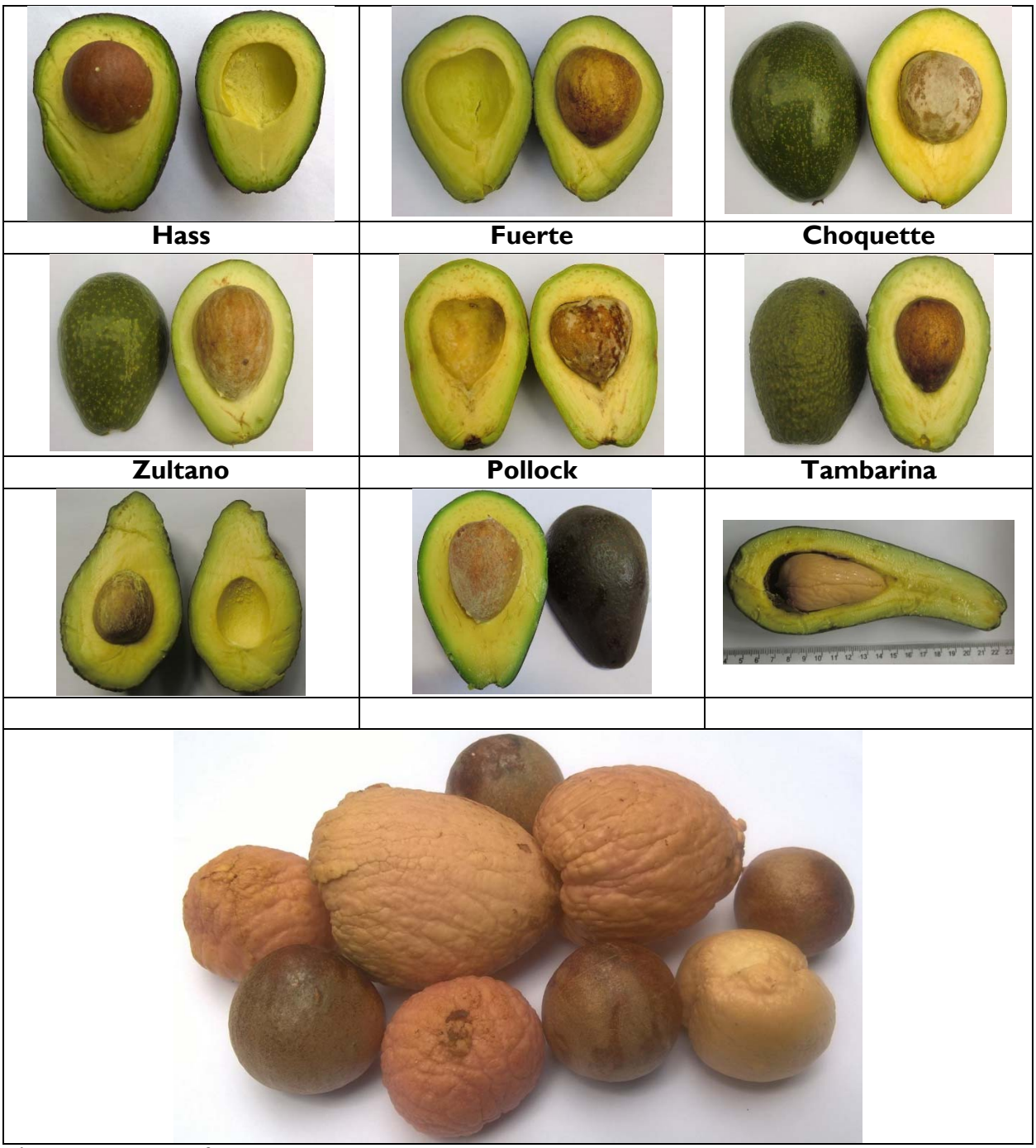


Fig. 1. Diversity of avocado in Kenya

1. Seedling Varieties of Avocado

Commercial varieties

Table 1. Commercially important avocado varieties in Kenya

Cultivar	Production	Characteristics
Hass (Guatemalan race) 	- It matures 8-9 months after blossoming - Produces medium sized oval shaped fruit (150 -300 g) - Peak flowering (October) peak harvest (June to September)	- Most popular for export in many parts of the world - Vigorous upright tree which propagates readily and is a consistent bearer - The fruit skin is green at early stages but darkens to a dark brown on maturity - Fruit skin is slightly pebbled, pear-shaped, thin and leathery - Hass has a high tolerance to anthracnose; good postharvest performance - Adapted to cooler locations and performs well in a monoculture.
Fuerte (a hybrid between Guatemalan and Mexican races) 	- Fruit matures 6-8 months after flowering - Peak flowering (September to October) peak harvest (March to May) - Produces a medium sized pear-shaped fruit (250 – 450 g) - The fruit has a high oil content (20-40%)	- A popular for export in many parts of the world - Fruits are green, slightly pebbled, pear shaped with good flavour - Grows into a large tree if not pruned - Has alternate bearing - Skin is thin and the seed is medium and conical - More susceptible to post harvest fruit rots than Hass - Variety is dominant in preparation of edible avocado oil in Kenya
Nabal (Guatemalan race) 	- Matures 8-9 months after flowering - Produces nearly round fruit that is medium to large size	- The tree is large - Fruit is green, smooth, almost spherical or slightly oval - Flavour is good - Bears in alternate years

Cultivar	Production	Characteristics
Source: https://www.specialtyproduce.com/produce/Nabal_Avocados_11226.php	- Has a low oil content of 10 to 15%	
Puebla (Mexican race) Source: https://www.specialtyproduce.com/produce/Puebla_Avocados_10594.php	- It matures 5-7 months after flowering - The fruit is medium in size - Fruit has an oil content nearly 20% - It matures 5-7 months after blossoming. The fruit is medium size; with an oil content nearly 20%.	- Spreads, dark green tree - Fruit is deep maroon purple, smooth, glossy and roundish, medium-sized conical seed - Flavour is good - In Kenya, fruits are sold locally - The seedlings are used as rootstocks for most cultivars

Source: Lusike Wasilwa

Rootstock Varieties

Table 2. Rootstock varieties for avocado in Kenya

Cultivar	Production	Characteristics
Puebla (Mexican race) Source: https://www.specialtyproduce.com/produce/Puebla_Avocados_10594.php	- This variety is commonly used as a rootstock for most cultivars - Has a deep root systems thus promoting resilience in water-stress periods	- Spreads, dark green tree. - Fruit is deep maroon purple, smooth, glossy and roundish, medium-sized conical seed. Flavour is good. - In Kenya, fruits are sold locally -
G755	- This is variety used as rootstock	- G755 is tolerant to Phytophthora root rot - The seedlings are used as a rootstock
Fuerte (a hybrid between Guatemalan and Mexican races)	- Fruit matures 6-8 months after flowering	- Used as a rootstock because a large area of Kenya is planted with Fuerte and thus the seed are readily available

Cultivar	Production	Characteristics
	- Peak flowering (September /October) peak harvest (March to May) - Produces a medium sized pear-shaped fruit (250 – 450 g) - Has a high oil content (20-40%)	- A popular for export in many parts of the world - Fruits are green, slightly pebbled, pear shaped with good flavour - Grows into a large tree if not pruned. - Has alternate bearing - Skin is thin and the seed is medium and conical - More susceptible than Hass to postharvest fruit rots

Source of pictures: Lusike Wasilwa

2. Nursery Preparation

Selection and Management of Mother Plants

Prior to seed collection, one needs to select and mark good mother trees to use as sources of high quality seed (Fig. 2). The major characteristics of a good mother tree are:

- Mature
- Free of pests and diseases
- Vigorous
- True-to-type
- High yielding

The mother plants should be maintained through regular watering, fertilization, pruning, and pest and disease management. It is strongly recommended that a nursery should have a mother block comprising popular plant varieties. For new mother blocks, obtain mother plants from certified nurseries.

Fig 2: Healthy avocado mother block (**Source:** Grace Watani)

Nursery Propagation

Avocado is propagated by grafting onto rootstocks that are resistant to *Phytophthora* root rot, are tolerant to salinity and calcareous soils, adaptability to soils, and resistant to lime induced chlorosis. Rootstocks selection is also based on preferred tree size.

Plant healthy egg-sized with the broad side down in to 6' x 9' tins, pots or polythene bags. Transplant the seedlings in boxes or seedbed with adequate drainage 3-4 weeks after germination in boxes or seedbed with adequate drainage. Water the seedlings until they attain a pencil size (3/4 cm) thickness at which stage they should be grafted (Fig. 2).

Soil media for transplanting should consist of 10 parts of top soil, 3 parts of manure, 3 parts of compost, 1-3 parts of sand at a ratio of 10:3:1-3 (v/v) and 300 g of compound fertilizer (17:17:17).

		Avocado seedling ready for grafting

Fig. 2. Steps for propagating avocado (Source: Lusike Wasilwa)

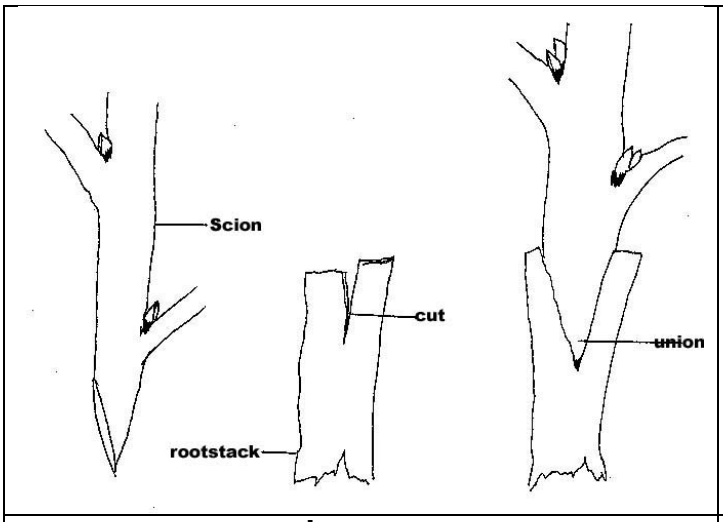
Grafting

Although top-wedge (Fig. 5) is the most suitable method for grafting, the following additional methods are also recommended. Grafting should be done at the point where the rootstock is soft and tender. The scion should be dormant at the time of grafting and should be of same size as the stock (Fig. 4). Wrap firmly with a polythene strap to exclude water from the graft union. Keep the grafted material under shade.

Secateurs	Pruning saw	Watering can
Grafting knife	Knife sharpening stones	Grafting tape

Fig. 3. Avocado propagation tools (Source: Lusike Wasilwa)

Top wedge or cleft grafting. Make a 3-cm wedge cut into the scion (Fig. 4a). Make a 3-cm cut into the rootstock (Fig. 4b). Ensure that all cut surfaces are smooth. Fit the scion and rootstock and align scion and rootstock cambium (Fig. 4c). Tie and wrap the graft union with plastic wrap. A plastic bag may be used to cover the scion and graft to prevent desiccation.



a b c

Fig. 4. Grafting avocado using the Top wedge method

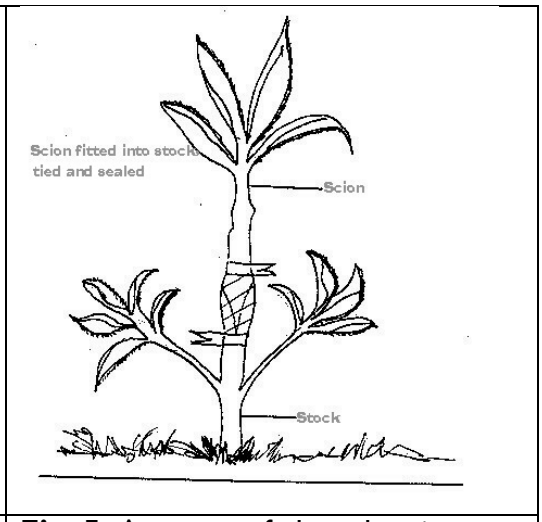


Fig. 5. Anatomy of plant showing scion, rootstock (stock)

Harvest and prepare scions when ready to graft	Prepare scion to expose cambium on two sides (1.5cm)	Scion show exposed cambium in v-shape
Cut a slit of 1.5 cm on the rootstock	Insert the scion into the rootstock with care	Note the v-shape at the graft union

Wrap the graft union with grafting tape	Ensure that the graft union is covered well to prevent loss of moisture	Remove the leave to reduce water loss
Place the grafted seedlings on raised ground that allows for proper drainage	Seedlings of avocado 4 weeks after grafting	Avocado seedlings under shade net
Successfully grafted seedlings	Avocado seedlings ready for transplanting	Use of wooden or cement panels to support avocado seedlings

Fig. 6. Propagation of avocado using top-wedge grafting (Source: Lusike Wasilwa)

Advantages of grafting

Grafted plants start to bear fruit 3 years earlier than un-grafted ones. They spread out, grow upright, are shorter and thus easier to harvest and r spray among other cultural activities. Grafted plants maintain the desirable characteristics of the mother plant such as fruit quality.

3. Soil, Water and Temperature Conditions

Soil Requirements

Under nursery conditions

- Soil is the most common growing medium for plants; therefore, good soil must be available within the nursery
- The soil used to raise seedlings should be fertile and should drain well and at the same time have capacity to retain moisture
- The site of soil collection should be clear from weeds, leaves and other debris
- The topsoil should be dug to a depth of about 10 cm

Fig.7: Top Soil (**Source:** Grace Watani)

Under field conditions

Soil analysis to determine the type and rate of fertilizer to be applied must be done before any fertilizer use recommendation is given. Quantity of manure and fertilizer application is dependent on the fertility status of the soil and the age of the trees. Nitrogen is the most important nutrient in avocado (Table 3) although other nutrients are also important (Table 3). At the onset of rain season remove mulch and scatter manure and fertilizer.

Water Requirements

- There must be enough water throughout the year as seedlings have to be watered twice or thrice a week depending on the weather.
- The quality of the water must also be considered since very few species can tolerate saline water.
- The distance between the water source and the nursery is important, since carrying water over long distances to the nursery is heavy and time consuming.
- Where there are water shortages, alternative water storage tanks should be installed.
- Earth dam can also be constructed for water harvesting and storage.
- Water requirements for the nursery stock will vary with the location and time of the year.
- Potted seedlings require about 5-10 mm per day and 15 mm in hot and dry areas. This is equivalent to 5-15 litres of water per sq. m. of a nursery bed.

Fig.8: Watering seedlings with hosepipe
(Source: Grace Watani)

Qualities of Good Nursery Media

- It should be light in weight for ease transport to the planting site
- Hold cuttings or seedlings firmly in place.
- Retain enough moisture to avoid need for frequent watering
- Be porous enough for excess water to drain easily.
- Allow sufficient aeration of the roots.
- Be free from un wanted seeds, nematodes, insect pests and diseases
- Be able to get effectively sterilized without changing its properties.
- Have enough nutrients for a healthy initial growth and development of seedlings.
- Have a suitable pH (5.5 to 6.5)
- Be stable and not swell or shrink excessively or crust-over under the sun (during hot seasons)

Permanent shade

- Involves huge investments for large nursery operators
- Mostly recommended for long-term investment
- A nursery should be constructed using durable materials

Fig. 9: Avocado nursery at KALRO-Thika demonstrating a permanent shade (**Source:** Eliud Gatambia)

4. Mulching and Weeding

Avocado is mulched to conserve moisture and also to add organic matter to the soil. Mulching therefore improves water retention of soil and subsequently promotes growth. The most common mulch of avocado is well-dried grass that is pest-free. Saw dust should be avoided as it will tie up nitrogen as it decomposes. Leaves may also be used as a mulch. Weeds around the trees should be maintained by cultivation. It is not recommended to use herbicides.

5. Fertilizer Application

Table 3. Fertilizer requirements per avocado tree

Age (years)	CAN (g)	DSP	Muriate of potash (g)	Manure (kg)
1-3	125	225	-	15
4-5	225	450	-	15
6-7	450	650	225	30
8-9	650	650	450	30
10-14	900	1000	635	-
15+	1300	1200	650	-

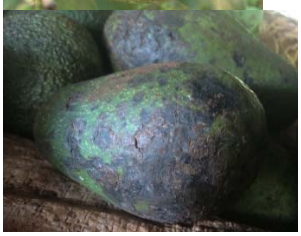
Table 4. Minor elements of importance in avocado

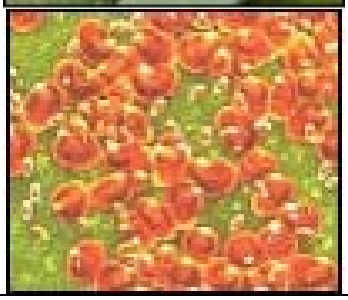
Deficiency	Symptoms	Remedy
Zinc	- Mottled leaves with light yellow areas between the veins and irregular development of shoots - In a severe stage, leaves and twigs becomes stunted and fruits misshapen - Tree growth is retarded and fruit production reduced - Fruits become round and not the usual pear shape	- Apply ($ZnSO_4$) at 0.5% and 0.25% Hydrated lime (Ca (OH) or 0.2% Zinc oxide or 250 g of Zinc sulphate for every year, up to maximum of 4.5 kg. The application should be done in a 60 cm strip around the drip of the tree
Manganese	- Low level of manganese cause leaf chlorosis	- Foliar spray of manganese sulphate when the leaves are young (i.e. just after a flush)
Iron	- Causes chlorosis of leaves	- Iron sulphate at the rate of 250 g per 10 litres of water. In acid soils Iron-Chelate at the rate of 360 g per tree
Chlorine	- Leaf scotch of the leaf (leaf-tip-burn) is caused by excess chlorine - Roots are damaged by high concentrations of chlorides	- Use a rootstock that is tolerant to saline condition

6. Pest & Disease Management

Table 5. Pests and Diseases of Economic Importance of Avocado

	Name of Pest/Disease	Symptom	Management Recommendation
	Diseases		
1	**Phytophthora root rot** (*Phytophthora cinammomi*) 	- *P. cinnamomi* is a root infecting and destroying pathogen -This leads to gradual debility of the above ground parts first noticed with leaf wilting and dieback of one or two branches on the tree -Yields decline gradually, leaves become smaller and more pale yellow-green, then drop -The twigs and branches die back	- **Cultural control -** Establish the orchard at a site where waterlogging conditions do not occur - Avoid standing water or did trenches to improve drainage - **Use of certified/clean planting material -** Planting disease-free and resistant/tolerant rootstocks - Maintaining good field and hygiene - Practice field sanitation and cultural activities - **Chemical control -** Soil drench with Ridomil (active ingredient metalaxyl) at the rate of 20-25g per square meter of canopy area. - **Chemical control -** Phosphonate (potassium phosphite). Note this is not the same as phosphoric acid (H_3PO_4). This chemical can be applied to the soil, to the foliage or, more commonly with avocado as a trunk injection - Apply 1500-3000 kg ha^{-1} of gypsum under the canopy and 6-12 inches of mulch 6 inches from the trunk
2	**Anthracnose** (*Colletotrichum gloeosporioides*) 	- Slightly sunken, brown to black circular spots on fruit - Spots (lesions) are first small and enlarge rapidly, under favourable conditions (wet or high humidity) becoming sunken and develop cracks from the centre	- **Chemical control -** Use copper based fungicides such as copper hydroxide and Kocide DF during fruit development. The PHI (postharvest interval) and REI (restricted-entry interval) for copper hydroxide are 0 day and 24 hours, respectively.

	Name of Pest/Disease	Symptom	Management Recommendation
		- The fungus can progress into the flesh, producing a greenish-black decay	
3	**Avocado scab** (*Sphaceloma persea*) 	- Scab mainly attacks fruit and foliage initially causing slightly raised brown to purplish-brown oval spots on fruit - As the fruit matures, the spots coalesce and their centres become sunken - Leaf lesions are less readily observed	• C • C • • **Chemical control –** • B •
4	**Cercospora spot** (*Cercospora purpurea*) 	- The fungus spots on avocado fruit and leaves - The disease is more prevalent on fruit during the rainy season - During the rainy season, grayish spore masses may occur on the surface of spots - Leaf spots may coalesce to form irregular areas of brown tissue	- C - C - - **Chemical control –** - B
5	**Avocado fruit rots**		- **Chemical control** - Use copper based fungicides such as copper hydroxide and Kocide DF during fruit development. The PHI

	Name of Pest/Disease	Symptom	Management Recommendation
	Fruit rot of Tambarina variety		(postharvest interval) and REI (restricted-entry interval) for copper hydroxide are 0 day and 24 hours, respectively.
	Pests		
1	**Mealybugs**	– Mealybugs suck sap from leaves, twigs, tender shoots, branches and fruits that causes crinkling, yellowing and rotting. – High infestation results to growth of sooty-mould on leaves, shots and fruits reducing photosynthetic area. – Severe infestations cause defoliation of leaves, reduce fruit set and increased dropping of young fruit.	- **Cultural control -** If the infestation is low, this pest can be controlled by rubbing from underside of leaves. - **Insecticidal soaps -** Can use spray with soapy solutions (1 to 2%) or insecticidal soaps (visit Pest Control Products Board for registered products). Spray only infested plants or plant-parts - **Oil control** - Use canola (rapeseed oil) vegetable oils or mineral oils. - **Botanical control** - Use botanicals e.g. neem products that function by repelling mealybugs
2	**Scales** Soft brown scale and Red Scale (*Aonidiella* spp.) 	- Soft scales suck sap from the foliage and in severe infestation cause complete distortion of the leaves with consequent drying and dropping off - They excrete large quantities of honeydew, supporting sooty-mold fungus. - Rarely appear on fruit	- **Oil control** - Scales can be controlled by using crop oils (DC – Tron Plus (Caltex oil). The oils should be made up as 1.5 to 3% solutions - **Insecticidal soaps** - Insecticidal soaps solutions (2%) are applied to completely cover the tree visit Pest Control Products Board for registered products) - **Chemical control -** Insecticides such as Diazinon maybe used to control scales as described by the manufacturer.

	Name of Pest/Disease	Symptom	Management Recommendation
3	**Systates weevil** (*Systates pollinosus*)	- This insect feeds on the leaves especially when the plants are young causing uneven edge on leaves - The insect feeds only at night and rests under the mulch or loose soil at the ground level during the day -	- **Cultural control** - The insects may be handpicked and destroyed - **Chemical control** - Dipterex, Sevin and Lebaycid as recommended by the manufacturer on the label.

(**Source of pictures:** Lusike Wasilwa except where indicated)

Important notice: Read the label on the pesticide container carefully and use the manufacturer's recommended rate. This applies to all the above mentioned pest and disease control products. Also visit **Pest Control Products Board** website for registered products on management of avocado pests and diseases.

7. Nursery Management

For a suitable nursery, site selection must be considered, including the location of the mother block, size of the nursery, source and quality of water, type of soil, topography i.e. slope and access. This determines the success or failure of a plant nursery. A good nursery will protect young trees and provide them with the best possible conditions for survival and growth.

Other factors in nursery management include:

- Consider whether seedlings are raised annually and the type of production system
- Whether seedlings are grown as bare-root, soilless mixture (pumice/coco peat) or in containers
- The space available for establishing the nursery
- Whether seedlings will be grown in pots or in beds
- Whether seedlings will be raised directly from seed, grafted, or from bare rooted cuttings
- The amount of water that is available to maintain seedlings

8. Transferring Seedlings from the Nursery

Transplanting

Seedlings should be planted into new ground (i.e. that does not have existing avocado). Avoid areas that have plants showing symptoms of Phytophthora root rot (see Table 5) should be avoided for new plantings

The planting holes should be 60 x 60 cm and should be dug during the dry spell. The topsoil and the subsoil should be kept separately. Fill the hole with the topsoil mixed with two *debes* of well-

decomposed manure and a handful of compound fertilizer such as NPK 20:10:10. Open the seedling containers carefully making sure that the soil covering the roots remains intact. Place the seedling in the centre of the hole and cover firmly with the with the mixture of top soil and FYM. Make a basin around the seedling for holding water (Fig. 10). ***Plant the seedlings at the same depth as it was in the nursery and water the plant immediately after planting.*** If the soil pH is below 4.5, 120 g $CaCO_3$ and 60 g single superphosphate (SSP) should be mixed with soil at each planting hole. Transplanting is more successful when carried out during the long rains either early morning or evening. In hot areas, shade the seedlings after transplanting.

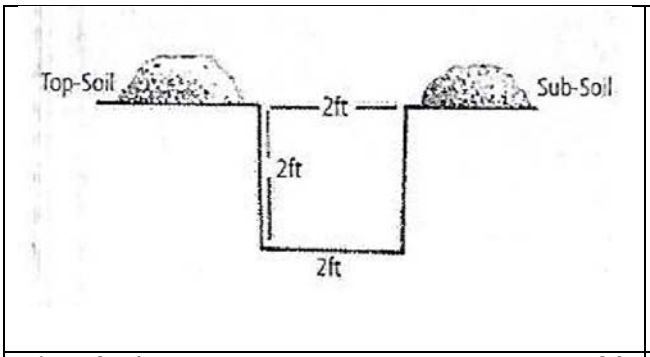


Fig 10. A hole to plant avocado should be 2ft wide x 2ft deep (or 60 x 60 cm)

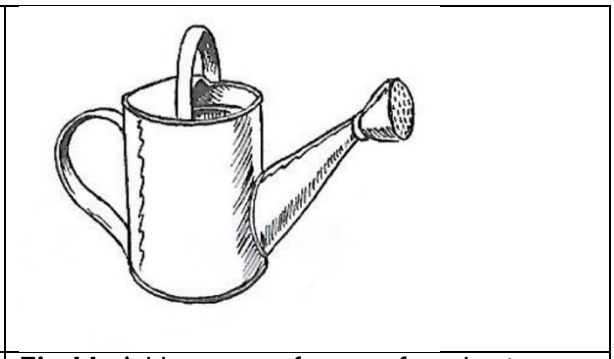


Fig 11. Add one can of water after planting

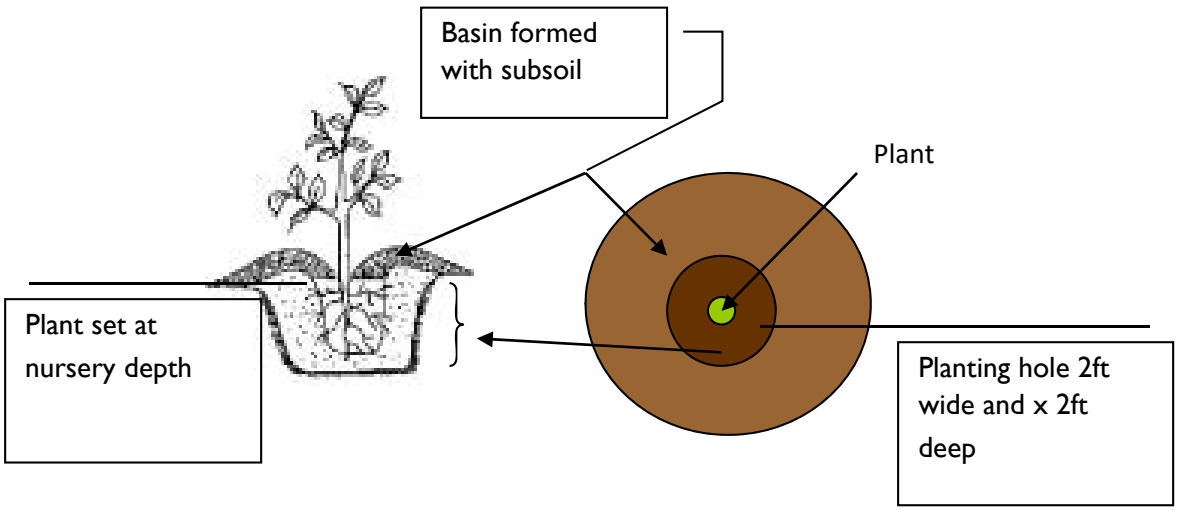


Fig. 11. How to plant avocado seedlings. The cross section further defines the layout of the hole.

9. Market

10. Seed Sources

In Kenya, avocado is commercially cultivated in Central, Lowe Eastern, Rift Valley, Western and Nyanza regions of the country. Planting material is sourced from KALRO, Agriculture Training Centres (ATC), Government of Kenya Prison Nurseries and private sector nurseries. This crop can be sourced indicated in the table below:

Table 6. Location of Nurseries Propagating Avocado

	County	Location	Contacts
		KALRO Centres	
1	Machakos	KALRO-Katumani	P. O. Box 340 – 90100, Machakos
2	Tharaka Nithii	KALRO-Marimanti	P. O. Box 27 - 60100, Embu
3	Taita Taveta	Wundanyi ATC	P.O. Box
4	Kakamega	KALRO-Kakamega	P. O. Box 169 - 50100, Kakamega
5	Busia	KALRO-Alupe	P. O. Box 278 – 50400, Busia, Kenya
6	Kisumu	KALRO-Kibos	P. O. Box 1490 – 40100, Kisumu
7	Kisii	KALRO-Kisii	P. O. Box 523 - 40200, Kisii
8	Nairobi	KALRO-Muguga South	P. O. Box 30148 - 00100, Muguga
9	Muranga	KALRO- NSRC	P. O. Box 7816 – 01000, Thika
10	Embu	KALRO-Embu	P. O. Box 27 – 60100 Embu
11	Muranga	KALRO-Practical Training Centre (PTC) or Macadamia Research Centre, Thika	P. O. Box 220 - 01000, Thika
12	Muranga	KALRO-Thika (Horticulture Research Institute)	P. O. Box 220 – 01000, Thika
13	Trans Nzoia	KALRO-Kitale	P. O. Box 450 - 30200, Kitale

	County	Location	Contacts
14	Nakuru	KALRO-Njoro	Private Bag – 20107, Njoro
15	Baringo	KALRO-Perkerra	P. O. Box 32 – 30403, Marigat
16	Kilifi	KALRO-Mtwapa	P. O. Box 25 – 80109, Mtwapa
17	Kilifi	KALRO-Msabaha	P.O. Box
18	Kwale	KALRO-Matuga	P. O. Box 4 – 80406, Matuga
		ATC	
1	Busia	Busia ATC	P. O. Box
2	Bungoma	Mabanga ATC	P. O. Box 33 50200, Bungoma
3	Taita Taveta	Kenyatta High School Mwatate	P. O. Box
4	Taita Taveta	TaitaTaveta University Campus (Ngerenyi);	P. O. Box
5	Taita Taveta	Taveta	P. O. Box
		GoK Prisons	
1	Kirinyaga	Kerugoya (G.K.) Prison	Farm Manager - 2034597845
2	Kirinyaga, Gathigiriri, Mwea	Mwea (G.K.) Prison	0721367578
3	Nakuru	Nakuru Prisons Nursery	Faith Ikara
4	Nyeri, Kingongo	Nyeri (G.K.) Prisons	0721974958 Gerald Muchiri Tel No.0723917922
5	Kisumu	Kibos (G.K.) Prisons	Mr. Okoth - 0720002080
6	Trans Nzoia, Kitale - Waitaluk	Kitale Annex Prison Nursery	0722448027
7	Nairobi	Kamiti Maximum Security Prison	P. O. Box
		Institutions	
1	Kiambu, Juja	JKUAT Enterprise Nursery	Francis Kimuyu - 0722333685
2	Kishaunet-Kapenguria	KVDA Suam Catchment (Kotoruk)	0727984016
3			
		Other Nurseries	
1	Bungoma - Naitiri	Good Neighbors	Zippy Simiyu -0722220889
2	Bungoma - Kanduyi-East Bukusu	Wamulu CBO	Robert - 0720931793; 0738500000
3	Busia County	Nuts and Oils International www.nutsandoilsinternational.com	Catherine – 0722-694-828
4	Eldoret, Kapsara	Benjamin Lukano	0733990574, 0725170692
5	Embu	Kamiru F. T. N	723780954
6	Embu	Karanja Njuguna nurseries	Karanja Njuguna - 0722334026
7	Homabay	Nyambok women group	Augustine Osunga
8	Homabay	Okosambu Farm Nursery	Okal (0727-209-267); Nasambu (0725-208-008)
9	Kirinyaga - Kibirigwi	Julia Nurseries	M. Njogu.
10	Kirinyaga - Kibirigwi	Kibirigwi Co-op Society	Nursery in-charge - 0710728451

	County	Location	Contacts
11	Kiriaini/Kangema	Mac-nut products	Mr. Mugi; 0722332496
12	Kisii	Akroplant Nursery	Silas Nyakundi – 0725-407-226
13	Kisii	Pamoja Nurseries	Joel Nyakundi – 0718-544-817
14	Laikipia central	Alime Commercial Nursery	723818160
15	Machakos District	ABC/ARTC Nursery	0722-859-579
16	Machakos District	Mwanzo Mpya Nursery	0714-326-227
17	Makueni District, Nzaui location	AIC Malooi Youth Nursery	0734-647-113
18	Mbeere North	Siakago Agro-forestry	Josephat Njeru - 0713021442
19	Meru, Marimanti	Umoja tree Nursery	0734771336
20	Meru	Mitunguu Irrigation Scheme	Patrick - 0712438698
21	Meru	Vision Growers	Julia - 0722895429
23	Meru, Nkubu	ICRAF/KAGURU FTC	Gitonga/
24	Muranga South - Nginda -Maragua	Gee Mac Agricultural Nurseries	Macharia Kabui Tel - 0722706037
25	Muranga South-Kariua	Mbari ya Mboche Nursery	Peter Gicheru Tel - 0721867625
26	Mwingi	Bidii Tree Nursery	James - 0722893148
27	Naivasha	Slopes Nursery - Paul Ndungu	0722342336
28	Nakuru Town	Mukulima Nursery - Joseph Opande	0725658913
29	Nakuru, Bahati	Friends Tree Nursery	Peter Githuka Njoroge
30	Nakuru, Subukia	Umoja Nursery - Joseph Mwangi	0724412461
31	Nanyuki	Sweet-waters Fruit Nursery	0723818160
32	Nyeri	Mahiga-ini Springs flowers & Fruit tree Nurseries	0701258645; 204756985
33	Nyeri, Tetu	Avocado Master Nursery	David Murigi; 0723367724
34	Nyeri / town	Erastus Ng'iru Ramuchi	712288596
35	Oloitok tok, Kimana	Ngaishi Environment Project - Parsanke Sayianka	psayianka@yahoo.com
36	Oloitok tok, Rombo	Tabby tree Nursery - Dorcas Wanjiku	0725345626
37	Siaya County		
38	Voi, Maungu	Wildlife Works Organic Project	Laurian Lenjo - 0722281851; Willy Kanyigi 0717574765

ATC – Agricultural Training Centre
KALRO - Kenya Agricultural & Livestock Research Organization
NRSC – National Sericulture Research Centre
PTC - Practical Training Centre

11. Social Media

- Facebook
- Twitter
- WhatsApp